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Mood computational mechanisms underlying increased risk behavior in adolescent suicidal patients

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eLife Assessment

This **valuable** study combined careful computational modeling, a large patient sample, and replication in an independent general population sample to provide a computational account of a difference in risk-taking between people who have attempted suicide and those who have not. It is proposed that this difference reflects a general change in the approach to risky (high-reward) options and a lower emotional response to certain rewards. Evidence for the specificity of the effect to suicide, however, is **incomplete**, which would require additional analyses.

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Abstract

Suicidal thoughts and behaviors (STB) rank among the foremost causes of death globally. While literature consistently highlights increased risk behavior in individuals with STB and identifies mood issues as central to STB, the precise cognitive and affective computational mechanisms driving this increased risky behavior remain elusive. Here, we asked 83 adolescent inpatients with affective disorders, where 58 patients with STB (S^+) and 25 without STB (S^-), and 118 gender/age-matched healthy control (HC) to make decisions between certain vs. gamble option with momentary mood ratings. Choice data analyses revealed more risk behavior in S^+ compared to S^- and HC. Using a prospect theory model enhanced with approach-avoidance parameters revealed that this rise in risky behavior resulted only from a heightened approach parameter in S^+. Furthermore, approach strength mediated the rise in gambling choices with STB severity. Altogether, model-based choice data analysis indicated dysfunction in the approach system in S^+ individuals, leading to greater propensity for gambling in favorable outcomes regardless the lotteries expected value. Additionally, mood

model-based analyses revealed reduced sensitivity to certain rewards in S^+ compared to S^- and HC. Importantly, these computational markers generalized to healthy population (n = 747). In S^+, mood sensitivity to certain reward was negatively correlated with gambling, offering a mood computational account for increased risk behavior in STB. These findings remained significant even after adjusting for demographic, clinical, and medication-related variables. Overall, our study uncovers the cognitive and affective mechanisms contributing to increased risk behavior in STB, with significant implications for suicide prevention.

Introduction

Every 40 seconds, a life is lost due to suicide(1). Suicidal thoughts and behaviors (STB) are one of leading causes of death worldwide that have devastating impacts on individuals, families, and societies. STB occurs from adolescence(2,3), especially in the context of mood disorders, e.g., major depressive disorder (MDD), anxiety disorder (AD), and bipolar disorder (BD)(4). Despite the progress made during the last 50 years for identifying risk factors (5) and developing preventing strategies(6), death rate from STB has not declined(7). The limited comprehension of cognitive and affective mechanisms creates a substantial gap in pinpointing targets for early prediction, screening, detection, and intervention in cases of suicidal thoughts. A recent and extensive literature has consistently reported increased risky decision making in patients with STB across different risk domains (for a short summary, see Table S1)(9–12). Yet, it remains whether this apparent increase in risk taking indeed corresponds to an increase in risk attitude, or a decrease in loss attitude or in value-independent choice biases, which can each result in a higher proportion of risky decisions. Understanding what is impaired in STB patients' decision process would be key to prevent STB, for example through cognitive behavioral therapy (13,14). However, why patients with STB adopt a riskier behavior and how decisions relate to mood dynamics remain unclear.

Although meta-analyses have shown increased risk behavior in patients with STB(9,11,12), the underlying cognitive computational mechanism is still unknown. Specifically, some studies found heightened loss aversion in STB in the context of the balloon analog risk task and the gambling task (15,16), while others observed the opposite pattern using the Iowa gambling task (17). Although all these results aligned with their hypotheses (for a short summary, see Table S1), this contradictory evidence may original from the use of underspecified models. A growing literature indeed shows that behavior can be far better explained after adding approach and avoidance components to prospect theory(18,19), that is by including a decision bias in favor of the highest gain (approach) and another decision bias against the lowest loss (avoidance), above and beyond options value difference. This class of models highlights the important role of motivational components in decision making in addition to traditional (expected) reward sensitivity (e.g., loss/risk aversion)(20). Importantly, STB has been proposed in theoretical work to result from abnormal motivational system(21–24), but no direct evidence support such proposals. Therefore, investigating motivational components may facilitate understanding why STB is associated with increased risk-taking behavior. We therefore hypothesized that heightened approach motivation, or weakened avoidance motivation, would account for increased risk behavior in STB.

While suicide is a decision process per se, atypical mood dynamics have been thought to be at the core of STB(3). Various suicidal-related theories, including the interpersonal theory(25), integrated motivational-volitional model(26), and three-step theory(27,28; see Figure S1 for a summary), have proposed that STB is initially caused by low mood experience. Some official organizations, e.g., National Institute of Mental Health, have also listed mood problems as warning signals(8). Interestingly, within the framework of decision making under uncertainty, gambling on lotteries with a revealed outcome has been found to induce high mood variance(29), providing an opportunity to assess the relationship between deficient mood and increased gambling decisions in STB. Specifically, in a gambling task with momentary mood ratings (also referred to happiness or subjective well-being), where participants were asked to make decisions between certain vs. gamble options (2 possible outcomes, 50% probability for each), Rutledge et.al., (2014) found that mood was sensitive to certain reward (CR), reward expectation (EV), and reward prediction error (RPE; the difference between experienced and expected outcome)(29). Although mood is thought to persist for hours, days, or even weeks(30–33), momentary mood, measured over the timescale in the laboratory setting, represents the accumulation of the impact of multiple events at the scale of minutes (30,32,34–38). Momentary mood external validity is demonstrated e.g., through its association with depression symptoms (37). By definition, mood is different from emotions, which reflect immediate affective reactivity and is more transient (e.g. from surprise to fear) (31–33,39). Here, we investigated which mood computational component (among CR, EV and RPE) is associated with STB. We expect the mood response to gamble related quantities (EV and RPE) to be lower in STB compared to the control groups. In contrast, riskier decision may result from aversion to certain reward in STB. Therefore, another possibility is that lower mood sensitivity to CR would relate to increased risk behavior in STB.

To summarize, the aim of this study is to examine cognitive and affective computational mechanisms underlying increased risk behavior in adolescent patients with STB, as adolescent period might provide a developmental window for opportunities for early intervention(2). This study aligns with the principles of Computational Psychiatry(40), which assumes that psychiatric symptoms arise from alterations in cognitive and affective computations. The ultimate aim for this field is to uncover "computational phenotypes"— distinct patterns of computational dysfunctions—potentially enabling targeted treatments, improved outcome predictions, and more precise diagnostic frameworks. Specifically, we employed a gambling task with momentary mood ratings to assess risk preferences and track mood fluctuations in response to various events. We applied computational models of risky decision making and momentary mood to dissect the cognitive and affective processes contributing to heightened risky behavior in patients with STB. Regarding choices, we hypothesized heightened approach motivation, or weakened avoidance motivation, in STB, which would account for increased risk behavior. Regarding mood dynamics, we hypothesized that greater mood sensitivity to gamble related variables (i.e., RPE and EV), or reduced mood sensitivity to CR, would explain increased risk behavior in STB.

Methods and materials

Participants

We recruited 95 adolescent patients with mood disorder from the Clinical Hospital of Chengdu Brain Science Institute, University of Electronic Science and Technology of China (The Mental Health Center of Chengdu, Sichuan, China). According to medical records and information from family and friends by the researcher (T.N) and psychiatrists (F.L, Y.Y, X.C, & Z.H), patients with suicidal thoughts and behaviors were categorized as suicidal group (S^+), while patients without suicidal thoughts and behaviors were identified as control group (S^-). The definition for suicidal thoughts in this study was active thoughts of suicide, i.e., wishing to die and having some intention to do so (see Supplementary Note 1 for details). This grouping operation was consistent with previous suicidal-related literature (41–46), reflecting the general tendency for suicidal risks, with important implications for suicidal prevention, especially among adolescence. As baseline control, we also recruited 124 gender- and age-matched healthy adolescents (HC). We assert that all procedures contributing to this work comply with the ethical standards of the ethical committee of The Clinical Hospital of Chengdu Brain Science Institute, University of Electronic Science and Technology of China (number: 2022(33)) on human experimentation and with the Helsinki Declaration of 1975, as revised in 2008. All procedures involving human subjects/patients were approved by the ethical committee of The Clinical Hospital of Chengdu Brain Science

Institute, University of Electronic Science and Technology of China (number: 2022(33)). Informed written consent was obtained. Patients were included if they met the following criteria: 1) both the researcher and psychiatrists agreed on their group classification; 2) they had a current diagnosis of major depressive disorder (MDD; unipolar depression), generalized anxiety disorder (GAD), or bipolar disorder with depressive episodes (BD), confirmed by two experienced psychiatrists using the Structured Clinical Interview for DSM-IV-TR-Patient Edition (SCID-P, 2/2001 revision; see Supplementary Note 1 for details);3) they were between 10 and 19 years of age; 4) they had experienced suicidal thoughts and behaviors during adolescence (ages 10-19); 5) they had no organic brain disorders, intellectual disability, or head trauma; 6) they had no history of substance abuse; 7) they had no experience of electroconvulsive therapy. In addition, participants were excluded if they failed more than 1/4 of the catch trials. The final sample consisted of 25 patients for S^-, 58 patients for S^+, and 118 HC participants. Note that the sample size of 25 in the control group was similar to previous suicidal- and risk decision-making-related research (47,49,51). See **Table 1** and Table S2 for demographic, clinical and psychological information. The validation dataset was from our previous online study, with 747 healthy participants completing the same task and numerous anxiety/depression-related questionnaires for different purposes. See (48) for demographic and psychological details.

Self-reported questionnaires

Participants completed a set of Chinese-version suicidal-, emotion regulation-, and depression/anxiety-related questionnaires. These measurements included the Beck Scale for Suicidal Ideation at the current time (BSI-C, 19 items) and at the worst time (BSI-W, 19 items) (50), the Childhood Trauma Questionnaire (CTQ, 28 items)(52), Emotion Regulation Questionnaire-Reappraisal (ERQ-R, 6 items) and Suppression (ERQ-S, 4 items)(53). In addition, as patients were available only for a limited duration, anxiety/depression-related scales from only 50 participants in the S^+ group and only 21 participants in the S^- group were collected. Specifically, patients filled the Trait subscale of the State-Trait Anxiety Inventory (TAI; 20 items)(54), the Penn State Worry Questionnaire (PSWQ; 16 items), the Beck Depression Inventory (BDI; 21 items) (55), and the Center for Epidemiologic Studies Depression Scale (CESD; 20 items)(56).

Experimental Procedure

Participants were asked to make a choice between a certain option and a gamble (50% probability for each outcome) to maximize their points and to rate their momentary moods(18,29). Before performing the task, participants were asked to rate their current happiness that we consider as their initial mood. At the beginning of the task, participants were endowed with 500 points. Each trial started with two options (a gamble option and a certain option) which were presented randomly on each side (**Figure 1A**). Upon response, the chosen option was highlighted in yellow for 0.5 s. Note that Rutledge et al., (2014) displayed the chosen option for about 6 s(29), a delay we shortened for the sake of time. Then the corresponding outcome at the screen center was presented for 1 s, followed by a fixation cross with a random duration (0.6~1.4 s). If the gamble was chosen, participants had equal probability to obtain each outcome. The obtained outcome was added to their total score, which was presenting at the top-right corner. Every 2~3 trials, participants rated their mood (“how happy are you at this moment?”) from 0 (very unhappy) to 100 (very happy) by moving a slider anchored at midpoint (i.e., 50). Upon identifying their current mood, a fixation cross was presented with a random duration (0.6~1.4 s). This task consisted of 90 randomly presented trials, including 30 mixed trials, 30 gain trials, and 30 loss trials. In mixed trials, participants made a choice between a certain amount 0 and a gamble with a gain amount {40, 45, or 75} and a loss amount determined by a multiplier {0.2, 0.34, 0.5, 0.64, 0.77, 0.89, 1, 1.1, 1.35, or 2} on the gain amount. For example, with a gain amount of 40 and a multiplier of 2 for the loss (2 times 40 = 80), participants chose between a certain option of 0 and a gamble option, which offered a 50% chance to win 40 and a 50% chance to loss 80. These trials are therefore particularly suited to measure loss aversion. In gain trials,

Table 1.

Demographics, clinical, psychological characteristics of patients with and without suicidal thoughts and behaviors.

	Group			Group contrast		S^+ vs. S^-	
	HC (n=118)	S^- (n=25)	S^+ (n=58)	F/χ^2	p	t/χ^2	*p*
Gender (female/male)	75/43	16/9	41/17	0.912	0.634	0.363	0.547
Age	15.31±2.15	15.68±1.75	14.83±1.80	1.868	0.157	**1.997**	**0.049**
BSI-C	1.29±3.62	2.84±2.66	18.02±7.56	224.230	<0.001	-9.754	<0.001
BSI-W	3.58±6.60	4.04±3.22	27.98±6.02	326.242	<0.001	-18.723	<0.001
CTQ	13.98±11.29	22.64±12.34	33.00±17.03	40.023	<0.001	-2.743	0.008
ERQ-R	14.77±4.38	13.08±6.34	8.48±5.39	31.317	<0.001	3.376	0.001
ERQ-S	6.86±3.64	8.80±3.77	10.79±3.79	22.322	<0.001	-2.200	0.031
Suicidal attempts history (yes)	---	---	29	---	---	---	---
Illness duration (months)	---	31.76±18.80	31.38±18.70	---	---	0.085	0.933
Family history (yes)	---	2	10	---	---	1.206	0.272
Current diagnosis (GAD/MDD/BD)	---	24/55/9	10/17/6	---	---	1.790	0.409
Medication (yes)	---	25	57	---	---	0.436	0.509
SSRI	---	16	39	---	---	0.082	0.775
SNRI	---	0	2	---	---	0.883	0.347
Trazodone	---	6	16	---	---	0.115	0.734
Antipsychotics	---	14	32	---	---	0.005	0.945
BZDs	---	20	45	---	---	0.060	0.807
Other anxiolytics	---	12	13	---	---	**5.434**	**0.020**
Mood stabilizer	---	13	18	---	---	3.282	0.070
TAI	43.49±8.54	50.38±12.19	65.36±7.62	108.863	<0.001	-6.276	<0.001
PSWQ	44.75±10.94	50.67±15.17	68.56±9.58	80.213	<0.001	-5.990	<0.001
BDI	9.45±9.43	18.62±15.11	38.30±9.67	129.516	<0.001	-6.573	<0.001
CESD	32.96±11.09	42.86±15.66	62.52±9.99	118.084	<0.001	-6.347	<0.001

Note: For the main results, we included gender, age, illness duration, family history, diagnosis, and various medications for control analysis. For anxiety/depression-related questionnaires (TAI, PSWQ, BDI, and CESD), due to time limitation, data from 8 participants in the S+ group and 4 participants in the S- group was not collected. Abbreviations: HC, healthy control; S^-, patients without suicidal thoughts and behavior; S^+, patients with suicidal thoughts and behavior; BSI-C, Beck Scale for Suicidal Ideation at the current time; BSI-W, Beck Scale for Suicidal Ideation at the worst time; CTQ, Childhood Trauma Questionnaire; ERQ-R, Emotion Regulation Questionnaire-Reappraisal; ERQ-S, Emotion Regulation Questionnaire-Suppression; AD, anxiety disorders; MDD, major depressive disorders; BD, bipolar disorders; SSRI, Selective Serotonin Reuptake Inhibitor; SNRI, serotonin-norepinephrine reuptake inhibitors; BZDs, Benzodiazepines; TAI, Trait Anxiety Inventory; PSWQ, Penn State Worry Questionnaire; BDI, Beck Depression Inventory, CESD, Center for Epidemiologic Studies Depression Scale.

there was a certain gain amount {35, 45, or 55} and a gamble with 0 and a gain amount determined by a multiplier {1.68, 1.82, 2, 2.22, 2.48, 2.8, 3.16, 3.6, 4.2, or 5} on the certain gain amount. In loss trials, there were a certain loss amount {−35, −45, or −55} and a gamble with 0 and a loss amount determined by a multiplier {1.68, 1.82, 2, 2.22, 2.48, 2.8, 3.16, 3.6, 4.2, or 5} on the certain loss amount. Many amounts and multipliers were used to provide a wide range of risk and loss sensitivity, as in previous literature. We also added an extra 4 trials in the entire task for attentional checks. For example, participants were asked to make a choice between a certain gain 20 and a gamble 35/55, where the correct response for this trial was the gamble choice (as the worst lottery outcome was higher than the certain reward). All experimental procedures were programmed using Psychopy3 (2021.2.3).

Choice computational models

In line with previous studies(18,19), our choice model space included expected value model (cM1), prospect theory model (cM2)(57), and approach-avoidance prospect theory model (cM3) (18). For cM2 (Equations 1-4), there were 3 parameters, including risk aversion (α, range: 0.3-1.3), loss aversion (λ: 0.5-5), and inverse temperature (μ : 0-10).

$$U_{gamble} = 0.5(V_{gain})^{\alpha} - 0.5\lambda(-V_{loss})^{\alpha} \tag{1}$$

$$U_{certain} = (V_{certain})^{\alpha} \ if \ V_{certain} \geq 0 \tag{2}$$

$$U_{certain} = -\lambda(-V_{certain})^{\alpha} \ if \ V_{certain} < 0 \tag{3}$$

$$P_{gamble} = \frac{1}{1+e^{-\mu(U_{gamble}-U_{certain})}} \tag{4}$$

where V_{gain} and V_{loss} are the objective gain and loss from a gamble, respectively. $V_{certain}$ is the objective value for the certain option. U_{gamble} and $U_{certain}$ are subjective utilities of the gamble and the certain option, respectively. Choice probability for gamble (P_{gamble}) is determined by the softmax rule.

Based on cM2, cM3 additionally considered motivational components when making decisions (Equations 1-3 & 5-8). That is, choice probability for P_{gamble} in cM3 is jointly determined by the softmax rule and approach/avoidance parameters (β_{gain}: - 1-1, β_{loss}: - 1-1).

For gain trials,

$$P_{gamble} = \frac{1-\beta_{gain}}{1+e^{-\mu(U_{gamble}-U_{certain})}} + \beta_{gain} \ \ if \ \beta_{gain} \geq 0 \tag{5}$$

$$P_{gamble} = \frac{1+\beta_{gain}}{1+e^{-\mu(U_{gamble}-U_{certain})}} \ \ if \ \beta_{gain} < 0 \tag{6}$$

For loss trials,

$$P_{gamble} = \frac{1-\beta_{loss}}{1+e^{-\mu(U_{gamble}-U_{certain})}} + \beta_{loss} \ \ if \ \beta_{loss} \geq 0 \tag{7}$$

$$P_{gamble} = \frac{1+\beta_{loss}}{1+e^{-\mu(U_{gamble}-U_{certain})}} \ \ if \ \beta_{loss} < 0 \tag{8}$$

Mood computational models

To quantify how different events impacted participants' momentary mood during the gambling task, we conducted a stage-wise model construction procedure(58). That is, we added or removed each component to the model progressively, based on the best model from the previous stage. In Stage 1, we fit the classic model assuming that momentary mood depends on the recency-

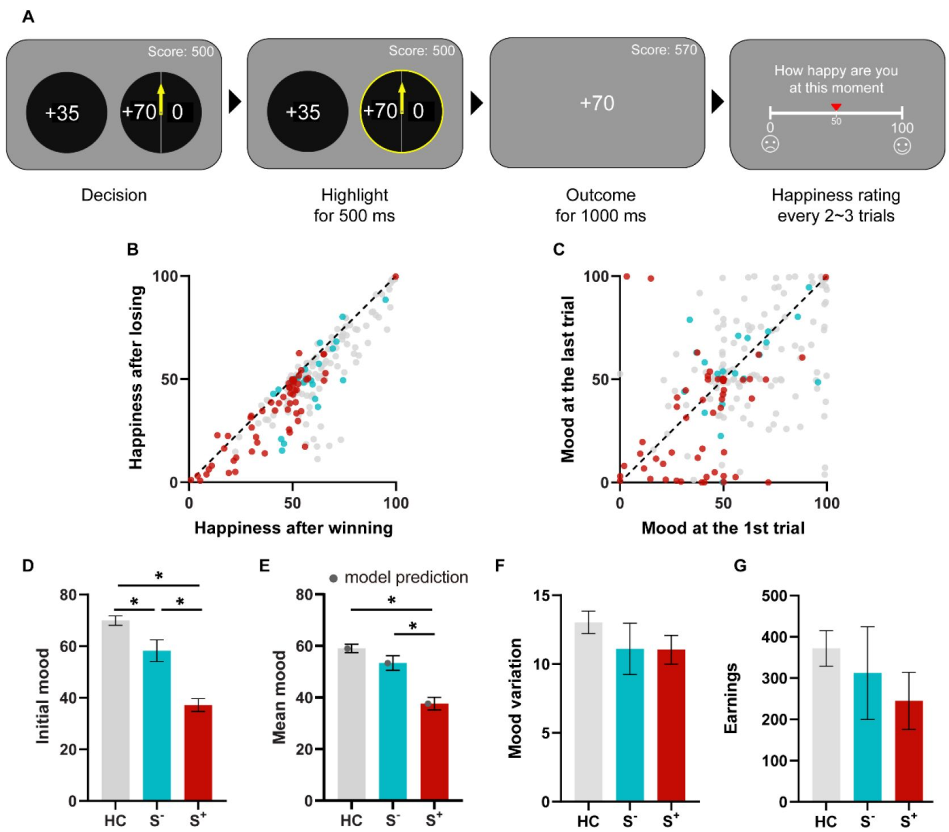


Figure 1.

Task design, outcome and time effects on mood, and group differences in mood.

A) Gambling task with mood ratings. On each trial, participants were asked to choose between a certain option and a gambling option (self-paced). Once selected, the chosen option was highlighted in yellow for 500 ms. Then the corresponding outcome was displayed in the center of the screen for 1000 ms. The cumulated score was always shown in the right-upper corner. Every 2 to 3 trials, participants were asked to complete a self-paced rating of their happiness, answering the question "How happy are you at the moment" on a slider from 0 (very unhappy) to 100 (very happy). B) Patients and healthy controls felt happier after winning than losing. C) Mood drifted over time. D) Group difference in mood before the task show weakened mood in S+. E) Group difference in average mood displays lower mood experience in S^{+}. The grey dots represent the winning model predictions. F) Mood variance was similar for all the three groups, as indexed by standard deviation of happiness ratings across the task. G) Each group earned about the same amount of point by the end of the task. Abbreviations: HC, healthy control; S^{-}, patients without suicidal thoughts and behavior; S^{+}, patients with suicidal thoughts and behavior; *p<0.05. Error bars correspond to the standard error.

weighted average of the chosen certain reward (CR), expected value of the chosen gamble (EV), and reward prediction error (RPE; mM1; Equation 9). RPE was defined as the difference between the obtained and expected value.

$$\text{Happiness}(t) = \beta_0 + \beta_{CR}\sum_{j=1}^{t}\gamma^{t-j}CR_j + \beta_{EV}\sum_{j=1}^{t}\gamma^{t-j}EV_j + \beta_{RPE}\sum_{j=1}^{t}\gamma^{t-j}RPE_j \quad (9)$$

Here, t and j are trial numbers, β_0 is a baseline mood parameter, other weights β capture the influence of different event types, $\gamma \in [0,1]$ is a decay parameter representing how many previous trials influence happiness. CR_j is the CR if the certain option was chosen on trial j; otherwise, CR_j is 0. EVj is the EV and RPEj is the RPE on trial j if the gamble was chosen. If the certain option was chosen, then $EV_j = 0$ and RPEj =0.

To check that mood ratings are best explained by a shared forgetting factor (i.e., the recency-weighted history of different event types), we compared a model with a single decay parameter to an alternative model, including a forgetting factors for each event type, e.g., different decay parameters for CR, EV, and RPE (mM2; Equation 10).

$$\text{Happiness}(t) = \beta_0 + \beta_{CR}\sum_{j=1}^{t}\gamma_{CR}^{t-j}CR_j + \beta_{EV}\sum_{j=1}^{t}\gamma_{EV}^{t-j}EV_j + \beta_{RPE}\sum_{j=1}^{t}\gamma_{RPE}^{t-j}RPE_j \quad (10)$$

Although mM1 has been shown to accurately predict mood data(29), we also fit alternative mood models. Firstly, we fit an alternative model in which mood ratings are explained by the recency-weighted average of the certain reward (CR) and the gamble reward (GR; mM3; Equation 11), a simple model providing a mood sensitivity parameter for certain rewards and gamble rewards. Secondly, we also fit a model with two forgetting factors, one for CR and one for GR (mM4; Equation 12).

$$\text{Happiness}(t) = \beta_0 + \beta_{CR}\sum_{j=1}^{t}\gamma^{t-j}CR_j + \beta_{GR}\sum_{j=1}^{t}\gamma^{t-j}GR_j \quad (11)$$

$$\text{Happiness}(t) = \beta_0 + \beta_{CR}\sum_{j=1}^{t}\gamma_{CR}^{t-j}CR_j + \beta_{GR}\sum_{j=1}^{t}\gamma_{GR}^{t-j}GR_j \quad (12)$$

In Stage 2, to identify whether mood can be better explained by different responses to better and worse gamble outcomes, we fit a model splitting GR into better and worse GR terms (mM5). We also fit a model with different decay parameters for each event based on mM5 (mM6).

In Stage 3, to check whether mood data can be better explained by a single event (CR or GR), we compared a CR-mood model (mM7) and a GR-mood model (mM8).

Model fitting and comparison

We fit model parameters by using the method of maximum likelihood estimation (MLE) with fmincon function of MATLAB (version R2015a) at the individual level. To avoid local minimum, we ran this optimization function with random starting locations 50 times. Bayesian information criteria (BIC) were used to compare model fits.

Replication of suicidal-related results in an independent dataset (n = 747)

We next verified our results in an independent dataset, including the same task and BDI questionnaire in 747 healthy participants (500 females; age: 20.90±2.41)(48). In particular, one item in BDI involves the measurement of STB. In item 9 of BDI, participants chose one option that describes them best: Option 1, “I don’t have any thoughts of killing myself.”; Option 2, “I have thoughts of killing myself, but I would not carry them out.”; Option 3, “I would like to kill myself.”; Option 4, “I would kill myself if I had the chance.”. We identified S^+ group as choosing Option 2, 3, or 4, while participants selecting Option 1 were categorized as S^- group. Therefore, there were 129 participants in S^+ and 618 participants in S^-. We did not find significant group difference in gender and age (*ps* > 0.075). To make it comparable, we fit the winning choice and mood models from the clinical study.

Statistical analysis

We performed chi-square, independent-sample t-test or repeated measure ANOVA to test group-related differences. Spearman correlations were used to check correlations among suicidal-related questionnaires, choice data, and mood data. Generalized linear model was conducted for control analysis using Matlab R2015a. Mediation analyzes were conducted using R (4.1.0) and the R package 'mediation'. All reported tests are two-tailed in addition to the replication of previous findings in the validation dataset. We set the significance level at p = 0.05. Multiple comparisons were corrected using Benjamini-Hochberg false discovery rate (FDR) correction.

Results

Demographic and clinical characteristics

Overall, gender and age were comparable among S^+, S^-, and HC groups (*ps* > 0.157), though S^+ was significantly younger than S^- (t = 1.997, p = 0.049). As expected, S^+ scored significantly higher than S^- and HC in suicidal-related scales (e.g., BSI-C; *ps* < 0.001), further validating our group manipulation. There was no significant difference between S^+ and S^- in illness duration, family history, diagnosis, and various medications use (*ps* > 0.07), except other anxiolytics (χ^2=5.434, p = 0.020). See **Table 1** and Table S2 for details. For subsequent control analysis for S^+ vs. S^- contrast, we included gender, illness duration, family history, diagnosis, and various medications use (except other anxiolytics) as covariates, whereas we used median split to check potential confounds of age and the use of other anxiolytics.

Sanity checks

To ensure engagement and task validation, we performed sanity checks. As expected, we found significant group differences in psychological measurements (*ps* < 0.001), including childhood trauma, emotion regulation, and anxiety/depression (**Table 1** and Table S2). In addition, we replicated the classic mood-related effects(59,60): 1) subjects were happier after winning than losing (t = 11.001, p < 0.001; **Figure 1B**) and 2) mood drifted over time (t = −3.254, p = 0.001; **Figure 1C**). As grouping checks, we found a hierarchical pattern of mood level both before the task and across the task ($S^+ < S^- <$ HC; for initial mood, F = 53.415, p < 0.001; S^+ vs. S^-: t = −4.525, p < 0.001; S^+ vs. HC: t = −10.427, p < 0.001; S^- vs. HC: t = −2.634, p = 0.009; **Figure 1D**; for mean mood, F = 28.018, p < 0.001; S^+ vs. S^-: t = −3.773, p < 0.001; S^+ vs. HC: t = −7.292, p < 0.001; S^- vs. HC: t = −1.458, p = 0.147; **Figure 1E**). No significant group difference in mood variation were found (F = 1.270, p = 0.284; **Figure 1F**) which suggests that any parameter difference between groups is unlikely to be explained by mood variance. Moreover, there was no group difference in terms of mood drift effect, or earnings (**Figure 1G**; *ps* > 0.276).

Choice results

To replicate previous findings of increased risk behavior in suicidal populations, we conducted a two-way ANOVA on gambling rate with group (S^+/S^-/HC) as a between-subject factor and trial type (mix/gain/loss) as a within-subject factor. We found a significant main effect of group (F = 3.655, p = 0.028, partial η^2 = 0.036; **Figure 2A**), with more gambling behavior for S^+ than S^- (two-sample t-test, t = 2.145, p = 0.035) and HC (t = 2.465, p = 0.015) and comparable gambling behavior between S- and HC (t = −0.439, p = 0.661) across the task. We also observed the main effect of trial type (F = 51.225, p < 0.001, partial η^2 = 0.206; gain > mix > loss). We did not observe any significant interaction effect between group and trial type (F =0.270, partial η^2 = 0.003). Within patients, this group effect on gambling rate remained significant after controlling for gender, illness duration, family history, diagnosis, and various medications use (*ps* < 0.05). There was also no significant age/other anxiolytics use difference in gambling behavior (*ps* > 0.109; Figure S9). In addition, there was significant correlations between gambling rate and Suicidal Ideation score at current time (BSI-C, rho = 0.233, p = 0.034; **Figure 2B**) and Suicidal Ideation score at worst time (BSI-W, rho = 0.219, p = 0.046) among patients.

We next performed a model comparison to select the model that best explain choice data. This analysis revealed that the winning model to formally quantify mechanisms for gambling behavior is the approach-avoidance prospect theory model (cM3; mean R^2 = 0.37; **Table 2**). Parameter and model recovery analyses showed that each model and parameter can be identified (rs > 0.91, *ps*< 0.001; see Supplementary Note 5; Figure S2 & S3). As predicted, we found a (marginally) significant group effect in approach parameter (F = 2.989, p = 0.053; **Figure 2C**), with a significant stronger approach motivation for S^+ than S^- (t = 2.217, p = 0.029) and HC (t = 2.091, p = 0.038), and comparable between S- and HC (t = −0.737, p = 0.463). No other significant group difference in these parameters was found (*ps* > 0.135). Within patients, this group effect on the approach parameter remained significant after controlling for gender, illness duration, family history, diagnosis, and various medications use (*ps* < 0.05). There was also no significant age/other anxiolytics use difference in gambling behavior (*ps* > 0.223; Figure S9). In addition, we observed significant positive correlations of approach parameter with BSI-C (rho = 0.286, p = 0.009) and BSI-W (rho = 0.222, p = 0.044) among patients, suggesting more gambling in gain (regardless of risk attitude) for patients with high STB severity. Given significant correlations between BSI-C, approach parameter, and gambling rate (*ps* < 0.034), we further conducted a mediation analysis with the assumption of the mediating effect of approach motivation of suicidality on the risk behavior. Results supported our hypothesis (a×b = 0.233, 95% CI = [0.074, 0.40], p < 0.001; **Figure 2D**). Taken together, these choice results suggest that suicidal thoughts and behavior increase risk behavior through stronger approach motivation.

Mood results

Next, we turned to mood model comparison. We observed inconsistent mood winning models for different groups (**Table 3**), suggesting an effect of STB on mood dynamics. Given that the focus of the current study was STB effect, especially for the S^+ group, with the baseline control of S^- and HC groups, we specially focused on the winning model from the S^+ group. Parameter and model recovery analyses showed that each model and parameter can be identified (rs > 0.90, ps < 0.001; see Supplementary Note 5; Figure S2 & S3). The winning mood model from S^+ assumed that momentary mood fluctuations were explained by the recency-weighted average of certain reward (CR) and the gamble reward (GR; mM3; mean R^2 = 0.42; **Table 3**). Overall, both CR and GR weights were significantly higher than 0 (CR: t = 8.033, p < 0.001; GR: t = 9.853, p < 0.001). The baseline parameter β_0 was significant correlated with the initial mood (rho = 0.580, p < 0.001), validating this model. We also replicated previous depression-related findings (61): depression symptom measured by Beck Depression Inventory (BDI) was negatively correlated with the baseline mood parameter β_0 (rho = −0.530, p < 0.001; Figure S5). We found significantly lower β_0 in S^+ than S^- (F = 22.861, p <0.001; t = −3.513, p < 0.001) and HC (t = −6.606, p < 0.001), which mirrors the lower initial mood pattern. Importantly, a two-way ANOVA on mood parameters with group (S^+/S^-/HC) as a between-subject factor, event type (CR/GR) as a within-subject factor showed a significant main effect of group (F = 3.835, p = 0.023, partial η^2 = 0.037), with lower mood sensitivity for S^+ than S^- (t = −2.080, p = 0.041) and HC (t = −2.758, p = 0.006) and comparable between S- and HC (t = −0.110, p = 0.913). We also observed a significant interaction effect between group and event type (F = 4.283, p = 0.015, partial η^2 = 0.041; **Figure 3B**). Simple effect analysis revealed that S^+ group exhibited significant lower mood sensitivity to CR as compared to GR (F = 4.823, p = 0.029, partial η^2 = 0.024), while there was no significant CR-GR difference in S^- (although trendy; F = 2.783, p = 0.097, partial η^2 = 0.014) and HC (F = 0.989, p = 0.321, partial η^2 = 0.005). This interaction was driven by the group difference in CR (F = 6.085, p = 0.003, partial η^2 = 0.058) rather

Figure 2.

Choice results.

A) Group differences in gambling behavior. The grey dots represent the winning model prediction. B) Positive correlation between Suicidal Ideation score at current time (BSI-C) and gambling behavior in patients. The lighter, semi-transparent dots represent individual participants, while the dark dot with an error bar indicates the mean of binned scores (for illustration purposes only). C) The estimated parameters from the winning choice model differed across groups. S+ exhibited stronger approach motivation than S- and HC. D) The mediation model among the BSI-C, β_{gain}, and gambling behavior in the gain condition among patients. The approach parameter mediated the effects of BSI-C on increased gambling behavior in the gain condition.

Abbreviations: HC, healthy control; S^-, patients without suicidal thoughts and behavior; S^+, patients with suicidal thoughts and behavior; BSI-C, Beck Scale for Suicidal Ideation at the current time; **p*<0.05.

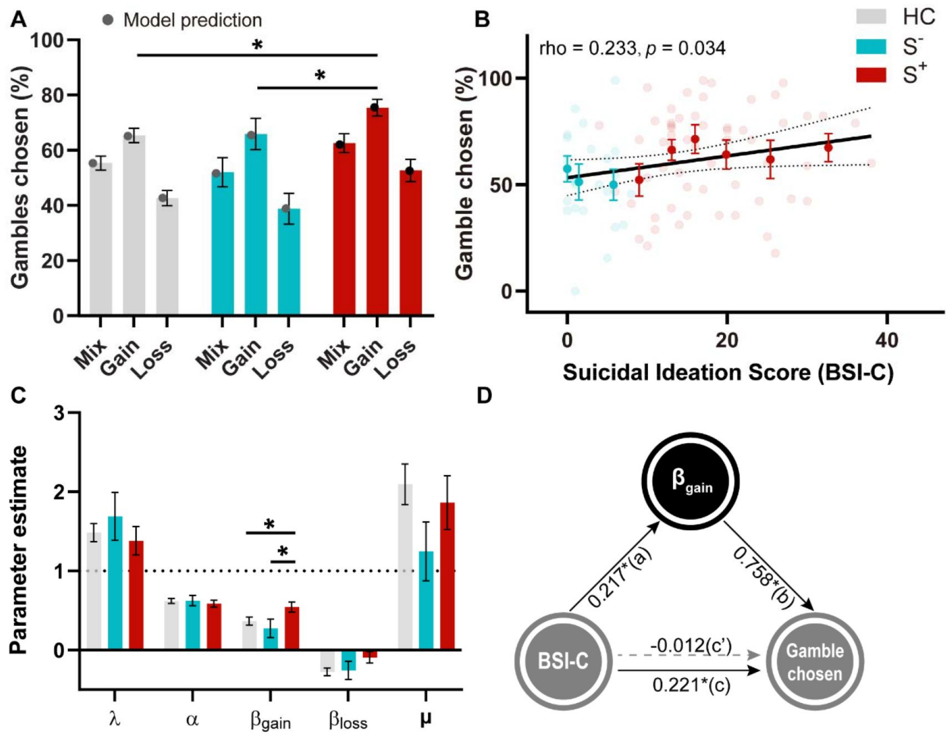


Table 2.

Choice model comparison.

Model #	Model specification	# of parameters	Δ BIC	meanR^2	Δ BIC for each group HC	S-	S+
1	μ	1	3873.16	0.08	2272.48	370.19	1230.49
2	λ, α, μ	3	3153.79	0.18	1822.07	263.97	1067.75
3	**λ, α, β_{gain}, β_{loss}, μ**	5	0	0.37	**0**	**0**	**0**

Abbreviations: Δ BIC, Bayesian information criterion relative to the winning model (cM3); HC, healthy control; S^-, patients without suicidal thoughts and behavior; S^+, patients with suicidal thoughts and behavior.

than in GR ($F = 0.801$, $p = 0.450$, partial $\eta^2 = 0.008$). Specifically, S^+ showed lower mood sensitivity to CR than S^- ($t = -2.661$, $p = 0.009$) and HC ($t = -3.381$, $p < 0.001$), while S^- and HC were comparable ($t = 0.450$, $p = 0.679$), suggesting S^+ was specifically more insensitive to certain outcome than gamble outcome. No significant main event type (CR vs. GR) effect was found ($F = 0.285$, $p = 0.594$, partial $\eta^2 = 0.001$). Within patients, this group effect on β_{CR} remained significant after controlling for gambling rate, earnings, mood-related outcome effect, mood drift effect, gender, illness duration, family history, diagnosis, and various medications use (*ps* < 0.032). There was also no significant age/other anxiolytics use difference in gambling behavior (ps > 0.582; Figure S9). In addition, we observed significant negative correlation between BSI-C and β_{CR} among patients ($rho = -0.243$, $p = 0.027$). These results indicate decreased mood sensitivity for certain reward in suicidal populations.

In addition to the winning model (mM3) from S^+ group, we also checked results from the classic mood model (mM1). Overall, we replicated previous findings (Figure S6): 1) mood sensitivity to CR, EV, and RPE were all significantly higher than 0 (*ps* < 0.001); 2) higher weight for RPE than EV ($t = 5.760$, $p < 0.001$). Although no significant group difference between S^+ and S^- was found in each parameter (*ps* > 0.115), we replicated significant correlation between BSI-C and β_{CR} ($rho = -0.239$, $p = 0.030$). To explore why the classic mood model (mM1) did not outperform the CR-GR model, we examined expectation effect on mood, as previous literature showed impaired value expectation in patients with STB(62). Our data suggests a lower mood sensitivity to RPE relative to EV in S^+ than HC (Figure S6; significant interaction between group and EV/RPE: $F = 3.422$, $p = 0.035$; with stronger mood sensitivity to RPE than EV in HC ($F = 36.658$, $p < 0.001$), while no such significant difference in S^+ ($F = 1.161$, $p = 0.283$) and S^- ($F = 3.009$, $p = 0.084$)). Equal weights on EV and RPE suggests that expectations cancel out as RPE is the difference between the outcome and EV, resulting in outcome only. Then, we additionally fit a mood model with CR, GR, and EV components (Figure S7). We expect less negative mood sensitivity to EV in S^+ than HC. As expected, in addition to replication of our main results (*ps* < 0.045; R^2 for this model: 0.487), we observed a less negative mood sensitivity to EV in S^+ than HC ($t = 2.302$, $p = 0.023$), which explains why the winning model shifts to mM3. Given that mM5 (splitting GR into better and worse terms) performed better than our winning model (mM3) in the S^- and HC groups, we also checked results from this model (Figure S8). Again, we found that S^+ had significant lower β_{CR} than S^- and HC (for group effect: $F = 44.660$, $p = 0.011$; S^+ vs. S^-: $t = -2.659$, $p = 0.009$; S^+ vs. HC: $t = -2.589$, $p = 0.010$; S^- vs. HC: $t = 1.059$, $p = 0.292$) and significant correlation between BSI-C and β_{CR}, ($rho = -0.297$, $p = 0.006$) among patients, suggesting the robustness of mood sensitivity to certain reward in suicidal people. The marginally significant group effect in approach parameter ($p = 0.053$) remain marginally significant after correction ($p=0.068$). In addition to this, all results of interest, including gambling chosen, approach parameter, and mood sensitivity to CR, remained significant with FDR correction (ps <=0.05).

Associations between choice and mood

To examine the association between risk behavior and atypical mood dynamics in suicidal patients, we then tested the correlation between participants' gambling rate and mood sensitivity to certain reward ($\beta_{CR)}$ in S^+. We found significant negative correlation between gambling rate and β_{CR} in S^+ ($rho = -0.274$, $p = 0.037$), suggesting the lower mood sensitivity to certain reward, the more gambling behavior suicidal patients made. We did not observe such a significant correlation in S^- ($rho = 0.246$, $p = 0.237$) and there was significant correlational difference between S^+ and S^- ($Z = -2.109$, $p = 0.017$; 42)), suggesting the suicidal-specific association of mood and choice.

Replication of suicidal-related results in an independent dataset (n = 747)

Next, we collected online data on healthy volunteers in an attempt to replicate our findings. In this large online dataset, we found lower mood experience in healthy volunteers who replied non-negatively to the Suicidal item of the BDI (S^+). Regarding the initial mood rating (before the task),

Table 3.

Mood model comparison.

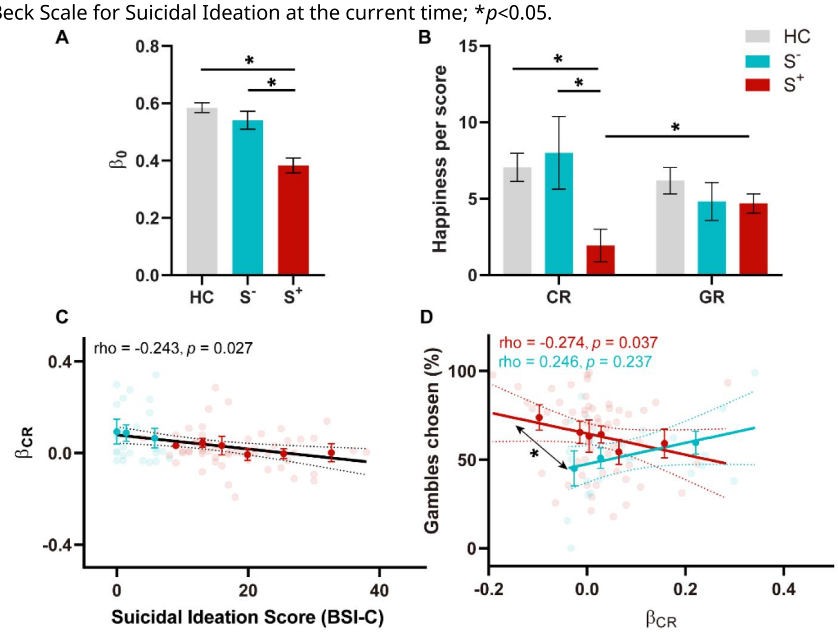
Model #	Model specification	# of parameters	Δ BIC	meanR^2	Δ BIC for each group HC	S$^-$	S$^+$
1	$\beta_0, \beta_{CR}, \beta_{EV}, \beta_{RPE}, \gamma$	5	-106.77	0.48	-182.04	32.54	42.73
2	$\beta_0, \beta_{CR}, \beta_{EV}, \beta_{RPE}, \gamma_{CR}, \gamma_{EV}, \gamma_{RPE}$	7	140.00	0.54	-69.40	83.20	126.20
3	$\boldsymbol{\beta_0, \beta_{CR}, \beta_{GR}, \gamma}$	4	0	0.42	0	0	**0**
4	$\beta_0, \beta_{CR}, \beta_{GR}, \gamma_{CR}, \gamma_{GR}$	5	-146.81	0.48	-272.15	26.37	98.97
5	$\beta_0, \beta_{CR}, \beta_{GR\_better}, \beta_{GR\_worse}, \gamma$	5	**-331.48**	0.49	**-355.10**	**-6.56**	30.18
6	$\beta_0, \beta_{CR}, \beta_{GR\_better}, \beta_{GR\_worse}, \gamma_{CR}, \gamma_{GR\_better}, \gamma_{GR\_better}$	7	-105.40	0.56	-313.09	81.34	126.34
7	$\beta_0, \beta_{CR}, \gamma$	3	2395.62	0.18	1379.96	264.87	749.79
8	$\beta_0, \beta_{GR}, \gamma$	3	403.46	0.34	228.69	21.24	153.52

Abbreviations: Δ BIC, Bayesian information criterion relative to the winning model in S$^+$ group (mM3); HC, healthy control; S$^-$, patients without suicidal thoughts and behavior; S$^+$, patients with suicidal thoughts and behavior.

Figure 3.

Effect of Suicidal thoughts and behavior on mood dynamics.

A) Group difference in mood baseline, β_0. B) Group differences in mood sensitivity to certain reward (CR) and gamble reward (GR). C) Correlation between Suicidal Ideation score at current time BSI-C and mood sensitivity to CR. D) Correlational difference in S- and S+ between mood sensitivity to CR and gambling behavior. Abbreviations: CR, certain reward; GR, gamble reward; HC, healthy control; S$^-$, patients without suicidal thoughts and behavior; S$^+$, patients with suicidal thoughts and behavior; BSI-C, Beck Scale for Suicidal Ideation at the current time; *p<0.05.



S^+ exhibited significantly lower mood than S^- (t = −6.077, p < 0.001; **Figure 4D**). There was a trend for lower mood experience across time in S^+ than S^- (t = −1.600, p = 0.055; **Figure 4E**). Critically, we identified a significantly increased gambling behavior in S^+ than S^-, especially in the gain domain (t = 1.668, p = 0.048; **Figure 4F**). Approach-avoidance prospect theory model (mean pseudoR^2 = 0.479) revealed a significantly heightened approach parameter in in S^+ than S^- (t = 1.762, p = 0.039; **Figure 4B**), but not any other choice parameters (ps > 0.172). We also replicated the previous mediation result that STB increase risk behavior through stronger approach motivation (a×b = 0.143, 95% CI = [0.016, 0.288], p = 0.031; **Figure 4C**). Regarding CR-GR mood model (mean R^2 = 0.588), we observed significantly lower β_0 in S^+ than S^- (t = −2.018, p = 0.022; **Figure 4F**). Mood sensitivity to CR (t = −2.237, p = 0.013; **Figure 4G**), but not GR (t = −0. 187, p = 0.473; **Figure 4G**), was significantly reduced in S^+ than S^-. These validation results suggest that our computational markers can generalize to healthy population. However, we did not observe any significant correlation between mood sensitivity to CR and gambling behavior (ps > 0.389), which suggests that the link between mood sensitivity to CR and gambling behavior may be specifically observable in suicidal patients.

Discussion

The current study tested cognitive and affective computational mechanisms for increased risk behavior in adolescent patients with suicidal thoughts and behaviors (STB), with a control group including adolescent patients without STB and gender/age-matched healthy control (HC). Firstly, we observed an increased gambling behavior and a lower overall mood in STB patients (S^+), as compared to non-STB patients (S^-) and HC, replicating previous findings(9–12). Secondly, using an approach-avoidance prospect theory model, we found heightened approach motivation in S^+ than S^- and HC, which explained increased gambling choices for STB, suggesting an over-reactivity of the approach system to approach risky options. Thirdly, using a momentary mood model, we showed that lower mood sensitivity to certain outcomes in S^+ compared to S^- and HC, which was driven by lower mood sensitivity to certain outcome in S^+ than S^- and HC. These computational markers generalized to healthy population (n = 747). Importantly, mood hyposensitivity to certain reward specifically correlated to more gambling behavior in S^+, offering a mood computational account for increased risk behavior in STB.

Our results suggest a unique reason for the twofold observations that STB patients display an increase in both risk taking and impulsivity, defined as a tendency to act quickly without planning while failing to inhibit a behavior that is likely to result in negative consequences(64–68). Indeed, we did not observe a difference in risk attitude per se between STB and controls but instead a higher approach behavior towards largest rewards (i.e., the lotteries) in STB patients. This would result from the value-independent term in the model that represent forms of approach in the face of gains(18,69,70). Such approach actions are elicited without regard to their actual contingent benefits and therefore corresponds to an impulsive behavior. A substantial body of research has shown that impulsivity, as assessed either through questionnaires or clinical observations, is a key predictor for STB (for a review, see Franklin et.al (2016)). Our study employed computational modeling to quantitively elucidate the altered approach-system processing for increased risky behavior in STB, offering enhanced predictive power and generalizability (40). On the other word, contrary to the proposal of atypical avoidance system(21,24), we did not observe significant group difference in avoidance, which may be attributed to the different involvement of the motivational system in learning and non-leaning contexts(19,71). In our model specification, motivational systems work in a value independent way in the non-learning context. Consistent with the view that suicide is an escape from intolerable affective states(3), risky behavior in suicidal individuals may be rewarding. In clinical practices, understanding the distortion of the approach system in STB may encourage mental health professionals to closely monitor patients who exhibit heightened approach

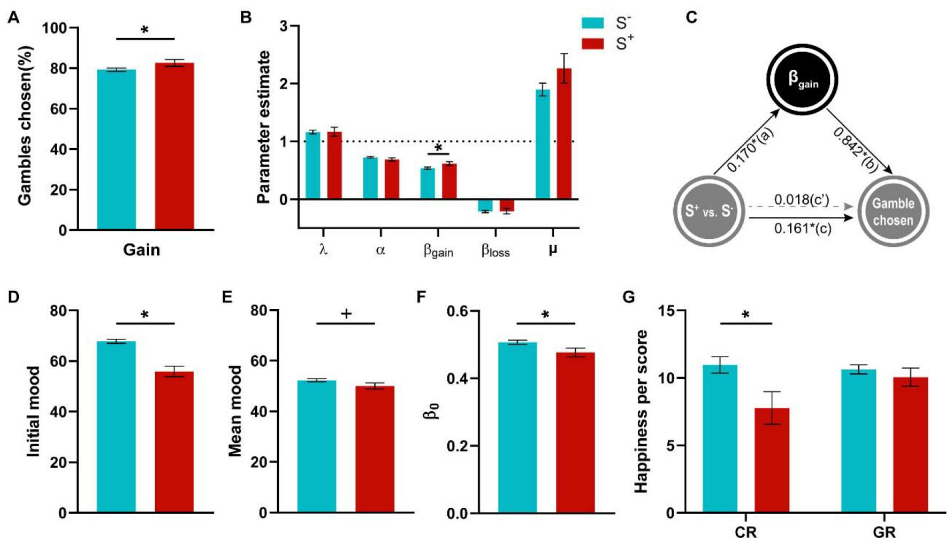


Figure 4.

Validation of suicidal-related results in an independent dataset of healthy populations (n = 747).

A) Group difference in gambling behavior in the gain domain. B) The estimated parameters from the winning choice model (pseudo R^2 = 0.479) differed across groups, with higher approach behavior for S^+. C) The mediation model among the group, β_{gain}, and gambling behavior in the gain condition. The approach parameter mediated the group effect on increased gambling behavior in the gain condition. **D)** Group difference in mood before the task show weakened mood in S^+. E) Group difference in average mood displays lower mood experience in S^+. FG) The estimated parameters from CR-GR mood model (mean R^2 = 0.588). F) Group difference in mood baseline, β_0. G) Group differences in mood sensitivity to certain reward (CR) and gamble reward (GR). Abbreviations: S^-, healthy participants without suicidal thoughts and behavior; S^+, healthy participants with suicidal thoughts and behavior; *p<0.05, $^+$ p<0.1.

tendencies. Such vigilance may enable early detection of risk-related behaviors, thus facilitating timely intervention strategies tailored to mitigate impulsivity-driven actions that may elevate the likelihood of STB.

Consistent with suicidal-related theories(3,72,73) and as summarized by Millner et.al (2020), we observed lower mood levels in patients with STB, regarding both initial happiness and mood baseline (the latter corresponding to the steady state mood converges to). More importantly, STB patients' mood was less sensitive to certain outcomes than control without STB, which would lead them to take more risk regardless the gain at stake and therefore to potentially experience more suboptimal outcomes than controls(9). Although no direct causal link was established between STB and happiness ratings in response to wins or losses, recent literature has documented associations between STB and anhedonia symptoms (albeit with mixed evidence; for a review, see (73)), where anhedonia can be assessed through affective reactivity to wins versus losses ((74)). Our findings thus provide support for the presence of anhedonia in STB, particularly in response to certain outcomes. Surprisingly, mood model-based analysis did not support the effect of expectations and prediction errors on mood in healthy people (the "CR-EV-RPE model"(18,29,75)), but suggest instead a dissociation between certain outcomes and lottery outcomes (the "CR-GR model"). These two models differed with respect to the inclusion of reward expectation terms, the former including it unlike the latter. This difference can be explained by the lower expected value signal in patients with STB(62), resulting in insufficient expectation representations of the gamble option to influence mood dynamics. An alternative explanation could be the duration of the chosen option display which was considerably lower in our design than in other mood studies (e.g., 0.5 s in our study versus 6 s in (29)), which would not let enough time for expectation to be built. Within the winning CR-GR model, we observed that S^+ specifically exhibited lower mood sensitivity to CR than GR, which was driven by mood hyposensitivity to CR in S^+ than S^- and HC. This mood insensitivity was associated with STB severity, which was replicated when using the CR-EV-RPE model. Importantly, we found that mood hyposensitivity to certain reward was specifically correlated to gambling behavior in patients with STB, suggesting the potential mood computational mechanism for increased risk behavior in STB. As for clinical practices, CR-based anhedonia linked to CR (computational reactivity) in STB may prompt mental health professionals to closely monitor patients who exhibit mood insensitivity to certain daily events. This proactive monitoring could aid in identifying and addressing risk-related behaviors early on.

With replication in an independent dataset with large sample size (n = 747), this study provides robust evidence of the affective and cognitive computational mechanisms underlying heightened risky behavior in adolescents with STB. In addition, these results remained significant after controlling for demographics, social and clinical variables, medication factors, and the timing of suicidal events (Supplementary Note 3 & 4). However, this study could not differentiate between suicidal thoughts and suicidal behaviors. Although it has been shown that they represented different decision-making processes with different neural underpinnings (76–78), our data did not reveal significant differences between them (see Supplementary Note 2). Future research would benefit from examining these distinctions at the neural level. Nonetheless, by combining the suicidal ideation and suicidal attempt groups into a single STB group (41–46), our findings highlight why adolescents with suicidality exhibit a preference for risky behavior, providing computational markers for general suicidal tendency among adolescents. These findings carry important clinical implications for early prevention of adolescent suicidality. Notably, this study, like many traditional studies on suicidality (41–43,79,80), does not seek to elucidate the affective and cognitive mechanisms underlying fluctuations in suicidal thoughts. Given the inherently variable nature of suicidal ideation, recent research has increasingly adopted ecological momentary assessments to capture real-time variations in suicidal ideations(46,81,82). While such methods can help predict when suicidal ideation may arise, they fall short of explaining the underlying mechanisms driving these thoughts. In contrast, our approach, consistent with traditional literature (41–45,79), is directed at understanding why individuals with STB are more inclined toward risky behavior. We acknowledge the interaction between environmental stressor and the occurrence of STB, noting that suicidal severity often diminishes once the stressor is removed (45,83). This is a crucially important issue in current psychiatric research. For instance, patients with MDD sometimes experience a depressive episode, particularly in response to stressful events. However, collecting data during STB is both impractical and ethically challenging. Our grouping approach is based on the assumption of trait-driven STB: individuals with a history of STB, despite not during the experiment, represent a cluster of suicidal-related traits (83,84). Sensitivity analyses for STB timeframe support this assumption (see Supplementary Note 3). We also recognize that these affective and cognitive impairments may worsen under stress(45,83). Future studies would benefit from investigating how acute stress influences the propensity for risky behavior in individuals with STB.

Given that STB is a challenging multifactorial phenomenon, the development of a formal theory to quantify suicide seems necessary(21,22,85). Our cognitive and affective computational insights may pave the way for such a formal theory. Although previous literature has shown various cognitive impairments(12), e.g., executive function, in STB(86), our work is the first to quantify mood dynamics impairment and their behavioral consequences, providing insight into potential target to prevent and intervene STB. Our results indeed provide a computational mechanism for the main theories of suicide, linking low mood to suicidal behaviors. Suicide behavior is conceived to result from an intention shaped by various motivational factors (e.g., feeling of entrapment, belongness, burdensomeness(87)). The suicidal intent may then progress to suicidal behavior, which is thought to be moderated by impulsive decisions (e.g., (88)). A possibility is that the approach component becomes excessive as the suicidal intent emerges. These findings provide new insights to the putative dynamics underpinning STB, and offer potential markers for the early prediction, screening, detection, and prevention of suicidal behavior. These results would explain the observed increase in risk-taking behavior in STB such as substance use, early onset of sexual intercourse and physical fighting independent of psychiatric diagnosis.

Several limitations are worth mentioning. First, although we found that aberrant mood sensitivity explained increased risk behavior in STB, the mutual relationship between mood and risk behavior remains to be tested. For example, does mood really influence risk behavior, does risk behavior influence mood, or is there a loop between them? Second, our cross-section findings are of correlational nature. Causal relationships remain to be tested in a longitudinal study.

To conclude, this study examined cognitive and affective computational mechanisms underlying increased risk behavior in adolescent patients with suicidal thoughts and behaviors. Given very limited predictive abilities of suicide from previous risk-factor investigations(8), our study offers a potential new perspective of mood, at the core of STB, and reveals a relationship between low mood sensitivity to certain reward and an increased risk behavior in STB and possibly suggesting dysfunctional dopaminergic and serotoninergic systems. Our work has important implications for prevention of suicide, especially for clinical populations.

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Additional information

Data Availability

All data produced in the present study are available upon reasonable request to the authors

Funding Statement

National Natural Science Foundation of China

Additional files

Supplementary materials.

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Reviewer #1 (Public review):

Summary:

The authors use a gambling task with momentary mood ratings from Rutledge et al. and compare computational models of choice and mood to identify markers of decisional and affective impairments underlying risk-prone behavior in adolescents with suicidal thoughts and behaviors (STB). The results show that adolescents with STB show enhanced gambling behavior (choosing the gamble rather than the sure amount), and this is driven by a bias towards the largest possible win rather than insensitivity to possible losses. Moreover, this group shows a diminished effect of receiving a certain reward (in the non-gambling trials) on mood. The results were replicated in an undifferentiated online sample where participants were divided into groups with or without STB based on their self-report of suicidal ideation on one question in the Beck Depression Inventory self-report instrument. The authors suggest, therefore, that adolescents with decreased sensitivity to certain rewards may need to be monitored more closely for STB due to their increased propensity to take risky decisions aimed at (expected) gains (such as relief from an unbearable situation through suicide), regardless of the potential losses.

Strengths:

(1) The study uses a previously validated task design and replicates previously found results through well-explained model-free and model-based analyses.

(2) Sampling choice is optimal, with adolescents at high risk; an ideal cohort to target early preventative diagnoses and treatments for suicide.

(3) Replication of the results in an online cohort increases confidence in the findings.

(4) The models considered for comparison are thorough and well-motivated. The chosen models allow for teasing apart which decision and mood sensitivity parameters relate to risky decision-making across groups based on their hypotheses.

(5) Novel finding of mood (in)sensitivity to non-risky rewards and its relationship with risk behavior in STB.

Weaknesses:

(1) The sample size of 25 for the S- group was justified based on previous studies (lines 181-183); however, all three papers cited mention that their sample was low powered as a study limitation.

(2) Modeling in the mediation analysis focused on predicting risk behavior in this task from the model-derived bias for gains and suicidal symptom scores. However, the prediction of clinical interest is of suicidal behaviors from task parameters/behavior - as a psychiatrist or psychologist, I would want to use this task to potentially determine who is at higher risk of attempting suicide and therefore needs to be more closely watched rather than the other way around (predicting behavior in the task from their symptom profile). Unfortunately, the analyses presented do not show that this prediction can be made using the current task. I was left wondering: is there a correlation between beta_gain and STB? It is also important to test for the same relationships between task parameters and behavior in the healthy control group, or to clarify that the recommendations for potential clinical relevance of these findings apply exclusively to people with a diagnosis of depression or anxiety disorder. Indeed, in line 672, the authors claim their results provide "computational markers for general suicidal tendency among adolescents", but this was not shown here, as there were no models predicting STB within patient groups or across patients and healthy controls.

(3) The FDR correction for multiple comparisons mentioned briefly in lines 536-538 was not clear. Which analyses were included in the FDR correction? In particular, did the correlations between gambling rate and BSI-C/BSI-W survive such correction? Were there other correlations tested here (e.g., with the TAI score or ERQ-R and ERQ-S) that should be corrected for? Did the mediation model survive FDR correction? Was there a correction for other mediation models (e.g., with BSI-W as a predictor), or was this specific model hypothesized and pre-registered, and therefore no other models were considered? Did the differences in beta_gain across groups survive FDR when including comparisons of all other parameters across groups? Because the results were replicated in the online dataset, it is ok if they did not survive FDR in the patient dataset, but it is important to be clear about this in presenting the findings in the patient dataset.

(4) There is a lack of explicit mention when replication analyses differ from the analyses in the patient sample. For instance, the mediation model is different in the two samples: in the patient sample, it is only tested in S+ and S- groups, but not in healthy controls, and the model relates a dimensional measure of suicidal symptoms to gambling in the task, whereas in the online sample, the model includes all participants (including those who are presumably equivalent to healthy controls) and the predictor is a binary measure of S+ versus S- rather

than the response to item 9 in the BDI. Indeed, some results did not replicate at all and this needs to be emphasized more as the lack of replication can be interpreted not only as "the link between mood sensitivity to CR and gambling behavior may be specifically observable in suicidal patients" (lines 582-585) - it may also be that this link is not truly there, and without a replication it needs to be interpreted with caution.

(5) In interpreting their results, the authors use terms such as "motivation" (line 594) or "risk attitude" (line 606) that are not clear. In particular, how was risk attitude operationalized in this task? Is a bias for risky rewards not indicative of risk attitude? I ask because the claim is that "we did not observe a difference in risk attitude per se between STB and controls". However, it seems that participants with STB chose the risky option more often, so why is there no difference in risk attitude between the groups?

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Reviewer #2 (Public review):

Summary:

This article addresses a very pertinent question: what are the computational mechanisms underlying risky behaviour in patients who have attempted suicide? In particular, it is impressive how the authors find a broad behavioural effect whose mechanisms they can then explain and refine through computational modeling. This work is important because, currently, beyond previous suicide attempts, there has been a lack of predictive measures. This study is the first step towards that: understanding the cognition on a group level. This is before being able to include it in future predictive studies (based on the cross-sectional data, this study by itself cannot assess the predictive validity of the measure).

Strengths:

(1) Large sample size.

(2) Replication of their own findings.

(3) Well-controlled task with measures of behaviour and mood + precise and well-validated computational modeling.

Weaknesses:

I can't really see any major weakness, but I have a few questions:

(1) I can see from the parameter recovery that the parameters are very well identified. Is it surprising that this is the case, given how many parameters there are for 90 trials? Could the authors show cross-correlations? I.e., make a correlation matrix with all real parameters and all fitted parameters to show that not only the diagonal (i.e., same data is the scatter plots in S3) are high, but that the off-diagonals are low.

(2) Could the authors clarify the result in Figure 2B of a correlation between gambling rate and suicidal ideation score, is that a different result than they had before with the group main effect? I.e., is your analysis like this: gambling rate ~ suicide ideation + group assignment? (or a partial correlation)? I'm asking because BSI-C is also different between the groups. [same comment for later analyses, e.g. on approach parameter].

(3) The authors correlate the impact of certain rewards on mood with the % gambling variable. Could there not be a more direct analysis by including mood directly in the choice model?

(4) In the large online sample, you split all participants into S+ and S-. I would have imagined that instead, you would do analyses that control for other clinical traits. Or, for example, you have in the S- group only participants who also have high depression scores, but low suicide items.

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Reviewer #3 (Public review):

This manuscript investigates computational mechanisms underlying increased risk-taking behavior in adolescent patients with suicidal thoughts and behaviors. Using a well-established gambling task that incorporates momentary mood ratings and previously established computational modeling approaches, the authors identify particular aspects of choice behavior (which they term approach bias) and mood responsivity (to certain rewards) that differ as a function of suicidality. The authors replicate their findings on both clinical and large-scale non-clinical samples.

The main problem, however, is that the results do not seem to support a specific conclusion with regard to suicidality. The S+ and S- groups differ substantially in the severity of symptoms, as can be seen by all symptom questionnaires and the baseline and mean mood, where S- is closer to HC than it is to S+. The main analyses control for illness duration and medication but not for symptom severity. The supplementary analysis in Figure S11 is insufficient as it mistakes the absence of evidence (i.e., $p > 0.05$) for evidence of absence. Therefore, the results do not adequately deconfound suicidality from general symptom severity.

The second main issue is that the relationship between an increased approach bias and decreased mood response to CR is conceptually unclear. In this respect, it would be natural to test whether mood responses influence subsequent gambling choices. This could be done either within the model by having mood moderate the approach bias or outside the model using model-agnostic analyses.

Additionally, there is a conceptual inconsistency between the choice and mood findings that partly results from the analytic strategy. The approach bias is implemented in choice as a categorical value-independent effect, whereas the mood responses always scale linearly with the magnitude of outcomes. One way to make the models more conceptually related would be to include a categorical value-independent mood response to choosing to gamble/not to gamble.

The manuscript requires editing to improve clarity and precision. The use of terms such as "mood" and "approach motivation" is often inaccurate or not sufficiently specific. There are also many grammatical errors throughout the text.

Claims of clinical relevance should be toned down, given that the findings are based on noisy parameter estimates whose clinical utility for the treatment of an individual patient is doubtful at best.

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Author response:

We thank the reviewers for recognizing the strengths of our work, as well as for their thoughtful and constructive feedback. In this provisional response, we focus on the main concern raised—namely, the need for stronger evidence that the effect is specific to suicide. A full revision of the manuscript will follow, in which we will address this point in greater depth and respond carefully to all additional comments in a point-by-point manner.

More specifically, reviewer 3 points out that “The main analyses control for illness duration and medication but not for symptom severity. The supplementary analysis in Figure S11 is insufficient as it mistakes the absence of evidence (i.e., p > 0.05) for evidence of absence.”. This is indeed an important point that we address below.

(1) Correction for symptom severity.

To address the request for evidence on specificity to suicidality beyond general symptom severity, we performed separate linear regressions to explain in gambling behaviour, value-insensitive approach parameter (β_{gain}), and mood sensitivity to certain rewards (β_{CR}) with group as a predictor (1 for S^+ group and 0 for S^- group) and scores for anxiety and depression as covariates. Results remained significant after controlling anxiety and depression (*ps* < 0.027).

Author response table 1.

	Gambling rate	β_{gain}	β_{CR}
Group	**β = 0.164, t = 2.305, *p* = 0.024**	**β = 0.374, t = 2.257, *p* = 0.027**	**β = -0.105, t = -3.461, *p* = 0.001**
TAI	β = -0.010, t = -1.796, *p* = 0.077	β = -0.008, t = -0.649, *p* = 0.519	β = 0.005, t = 1.921, *p* = 0.059
PSWQ	β = -0.001, t = -0.401, *p* = 0.690	β = -0.008, t = -0.968, *p* = 0.337	β = -0.002, t = -1.282, *p* = 0.204
BDI	β = 0.002, t = 0.519, *p* = 0.606	β = -0.003, t = -0.272, *p* = 0.787	β = -0.004, t = -2.302, *p* = 0.025
CESD	β = 0.006, t = 1.585, *p* = 0.118	β = 0.015, t = 1.652, *p* = 0.103	β = 0.003, t = 1.942, *p* = 0.056

Given high correlations among anxiety and depression questionnaires (rs > 0.753, *ps* < 0.001), we performed Principal Components Analysis (PCA) on the clinical questionnaire to extract the orthogonal components, where each component explained 86.95%, 7.09%, 3.27%, and 2.68% variance, respectively. We then performed linear regressions using these components as covariates to control for anxiety and depression. Our main results remained significant (*ps* < 0.027).

Author response table 2.

	Gambling rate	β_{gain}	β_{CR}
Group	**β = 0.164, t = 2.305, *p* = 0.024**	**β = 0.378, t = 2.257, *p* = 0.027**	**β = -0.105, t = -3.461, *p* = 0.001**
PC1	β = -0.007, t = -0.425, *p* = 0.672	β = -0.010, t = -0.247, *p* = 0.806	β = 0.009, t = 1.169, *p* = 0.247
PC2	β = -0.073, t = -1.569, *p* = 0.122	β = -0.164, t = -1.509, *p* = 0.136	β = -0.011, t = -0.532, *p* = 0.596
PC3	β = 0.098, t = 1.429, *p* = 0.158	β = 0.205, t = 1.283, *p* = 0.204	β = 0.037, t = 1.255, *p* = 0.214
PC4	β = -0.086, t = -1.133, *p* = 0.262	β = 0.010, t = 0.054, *p* = 0.957	β = 0.086, t = 2.657, *p* = 0.010

We believe that these analyses provide evidence that the main effects on gambling and on mood were specific to suicide.

(2) Evidence of absence of effect of symptom severity

Based on clinical interviews, we included patients with and without suicidality (S^+ and S^- groups). However, in line with suicidal-related literature (e.g., Tsypes et al., 2024), S^+ and S^- differed substantially in the severity of symptoms (see Table 1). Although we median-split patients by the scores of general symptoms (e.g., depression and anxiety) and verified no significant differences in these severities (Figure S11), the "absence of evidence" cannot provide insights of "evidence of absence". We, therefore, additionally conducted Bayesian statistics in gambling behavior, value-insensitive approach parameter, and mood sensitivity to certain rewards. BF_{01} is a Bayes factor comparing the null model (M_0) to the alternative model (M_1), where M_0 assumes no group difference. BF_{01} > 1 indicates that evidence favors M_0. As can be seen below, most results supported null hypothesis, suggesting that general symptoms of anxiety and depression overall did not influence our main results.

Author response table 3.

BF_{01}	TAI	PSWQ	BDI	CESD
Gambling rate	4.080	2.768	2.425	0.820
β_{gain}	3.819	3.911	2.987	1.128
β_{CR}	3.704	3.826	1.185	3.178

Overall, we believe that these analyses provide compelling evidence for the specificity of the effect to suicide, above and beyond depression and anxiety.

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